SSC MTS 2024 question paper with answers

4 November 2024 · Shift 2

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

During the first year, the population of a village increased by 5%, and during the second year, it reduced by 6%. At the end of the second year, its population was 35,532. What was the population at the beginning of the first year?

(A) 36,000
(B) 38,000
(C) 37,568
(D) 34,567

Q.3

Determine the smallest number which when divided by 20, 15 and 30 leaves remainder 6 in each case.

(A) 56
(B) 66
(C) 96
(D) 36

Q.4

A car's radiator should contain a solution of 52% antifreeze. A motorist has 8 litres of 40% antifreeze. How many litres (rounded off to 2 decimal places) of 95% antifreeze liquid should he add to his solution to produce a 52% antifreeze?

(A) 3.45
(B) 2.23
(C) 1.92
(D) 4.26

Q.5

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 3% per year rate of interest for 5 years.

(A) 390
(B) 450
(C) 410
(D) 490

Q.8

A cyclist covers a distance of 2.5 km in 4 minutes 10 seconds. How long will he take to cover a distance of 12 km at the same speed?

(A) 12 minutes
(B) 10 minutes
(C) 11 minutes
(D) 20 minutes

Q.9

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 69 kg and 35 kg is:

(A) 50.8
(B) 49.8
(C) 48.8
(D) 51.8

Q.10

The outer circumference of the top of a large hemispherical iron bowl is 565.2 cm and the bowl is y cm thick. If the total surface area of the bowl is 98,988.5 cm^2, then what is the value of y? (Use $\pi = 3.14$.)

(A) 4
(B) 3.5
(C) 5
(D) 6

Q.11

If a sum doubles itself on certain rate of simple interest in 10 years, then at the same rate of interest, the sum will become 4 times itself in:

(A) 30 years
(B) 40 years
(C) 25 years
(D) 20 years

Q.12

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 5% per year rate of interest for 3 years.

(A) 410
(B) 390
(C) 450
(D) 490

Q.14

A man on a tour travels first 120 km at 48 km/h and the next 120 km at 80 km/h. The average speed for the first 240 km of the tour is:

(A) 66 km/h
(B) 64 km/h
(C) 60 km/h
(D) 72 km/h

Q.15

A trader gains 44% after selling an item at 25% discount on the printed price. What is the ratio of the cost price and the printed price of the item?

(A) 35 : 48
(B) 48 : 35
(C) 48 : 25
(D) 25 : 48

Q.16

In a school, class 11th has three sections A, B and C. There are 30 students in Section A, 36 students in Section B, and 40 students in Section C. The average marks scored by the students in Section A, Section B and Section C in the Mathematics annual exam are 90, 85 and 70, respectively. Find the average marks scored by the whole class, correct to two places of decimals.

(A) 58.70
(B) 80.57
(C) 80.75
(D) 58.07

Q.19

The resistance of a wire is directly proportional to its length and inversely proportional to the square of its radius. Two wires of the same material have the same resistance and their radii are in the ratio of 7 : 8. The corresponding lengths of the wires are 147 cm and L cm, respectively. Find the value of L.

(A) 192
(B) 158
(C) 128
(D) 168

Q.20

A shopkeeper mixes three types of seeds worth ₹240, ₹258 and ₹252 in 3 : 1 : 2. Average cost of the mixture is:

(A) ₹248
(B) ₹247
(C) ₹245
(D) ₹246

Reasoning Ability and Problem Solving 15 questions

Q.1

E, F, G, H, L, K, and M are sitting around a circular table, facing the centre (but not necessarily in the same order). K sits to the immediate left of F. Only two people sit between K and L when counted from the left of K. Only two people sit between M and E when counted from the right of E. E sits to the immediate right of H. H is not an immediate neighbour of F. Who is sitting to the immediate right of G?

(A) H
(B) L
(C) M
(D) F

Q.2

In a code language, 'STATION' is coded as '18' and 'JOURNALS' is coded as '20'. How will 'GOLDMEDAL' be coded in the same language?

(A) 22
(B) 72
(C) 28
(D) 44

Q.4

In a certain code language, ‘system is backbone’ is written as ‘sdf wer tce’ and ‘backbone is you’ is coded as ‘tce kpo wer’. What is the meaning of the code ‘sdf’ in the given language?

(A) is
(B) backbone
(C) system
(D) you

Q.6

What will come in the place of the question mark (?) in the following equation, if ‘+’ and ‘×’ are interchanged and ‘÷’ and ‘–’ are interchanged? 35 ÷ 4 + 6 – 3 × 12 = ?

(A) 28
(B) 39
(C) 35
(D) 29

Q.7

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts,decide which of the given conclusions logically follow(s) from the statements. Statements: All roses are lilies. No rose is a sunflower. Conclusions: (I) All lilies are roses. (II) No lily is a sunflower.

(A) Only conclusion I follows
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Both conclusions I and II follow

Q.8

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: All bags are duffles. All duffles are containers. Conclusions: (I) Some containers are duffles. (II) All bags are containers.

(A) Neither conclusion (I) nor (II) follows.
(B) Only conclusion (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (I) follows.

Q.9

What will come in the place of the question mark (?) in the following equation if ‘+’ and ‘–’ are interchanged and ‘×’ and ‘÷’ are interchanged? 12 × 3 – 5 ÷ 4 + 11 = ?

(A) 13
(B) 14
(C) 11
(D) 12

Q.10

In a certain code language, ‘SNOW’ is written as ‘7358’ and ‘BLEW’ is written as ‘1293’. How is ‘W’ coded in the given language?

(A) 8
(B) 1
(C) 9
(D) 3

Q.11

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). Only 3 people sit between C and E when counted from the right of C. Only 1 person sits between E and G when counted from the left of E. F sits immediate to the left of D. Only 2 people sit between B and F when counted from the left of B. How many people sit between G and B when counted from the left of G?

(A) 1
(B) 3
(C) 4
(D) 0

Q.12

What should come in place of the question mark (?) in the given series? 41, 48, 55, 62, 69, ?

(A) 74
(B) 75
(C) 76
(D) 77

Q.14

BMGC is related to GRDZ in a certain way based on the English alphabetical order. In the same way, LWAW is related to QBXT. To which of the following is VGUQ related, following the same logic?

(A) BKRO
(B) ZKSM
(C) ALRN
(D) BMQN

Q.16

11 is related to 106 following a certain logic. Following the same logic, 15 is related to 146. To which of the following is 19 related following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 189
(B) 178
(C) 177
(D) 186

Q.17

MK14 is related to QO19 in a certain way. In the same way, XV31 is related to BZ36. To which of the following is RP52 related, following the same logic?

(A) ED43
(B) ST58
(C) VT57
(D) RT90

Q.18

What should come in place of the question mark (?) in the given series based on the English alphabetical order ? DBF, HFJ, LJN, PNR, ?

(A) FDH
(B) QOS
(C) TRV
(D) SQU

Q.19

Select the set in which the numbers are related in the same way as are the numbers of the given sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (284, 238) (271, 225)

(A) (212, 166)
(B) (185, 135)
(C) (297, 246)
(D) (252, 196)

General Awareness 24 questions

Q.1

At which of the following places is the tomb of Sher Shah Suri situated?

(A) Bargaon
(B) Rajgir
(C) Sasaram
(D) Sultanpur

Q.2

After the decisive battle of Plassey of 1757, who among the following was declared as the Nawab of Bengal by the British?

(A) Murshid Quli Khan
(B) Mansur Ali Khan
(C) Mir Qasim
(D) Mir Jafar Ali Khan

Q.3

Which of the following countries won its 8th Men's Asian Cup cricket title in 2023?

(A) Sri Lanka
(B) Bangladesh
(C) Pakistan
(D) India

Q.4

Which of the following dance awards is organised by the National Institute of Indian Classical Dance?

(A) Abhinandan Saroja
(B) Sanatan Sangeet Sanskriti
(C) Natya Shiromani National
(D) Bharata Shastra Nritya Praveena

Q.5

Ranjani Hebbar was associated with which of the following fields?

(A) Bhajan
(B) Carnatic music
(C) Hindustani music
(D) Folk music

Q.6

Which element is known for having the highest electronegativity on the periodic table?

(A) Carbon
(B) Oxygen
(C) Fluorine
(D) Hydrogen

Q.7

What does the term 'deuce' signify in tennis?

(A) A fault in serving
(B) The final point in a game
(C) A tied score at 40-40
(D) A let

Q.8

Which virus is responsible for causing common cold?

(A) Influenza
(B) Hepatitis A
(C) Rhinovirus
(D) Coronavirus

Q.9
In which of the following years was the 11th Fundamental Duty added in the Indian Constitution through the 86th Amendment Act?

(A) 1986
(B) 1982
(C) 1976
(D) 2002

Q.10
In the year 1930, the Salt Satyagraha was led by who among the following?

(A) Subhas Chandra Bose
(B) Mahatma Gandhi
(C) Sardar Vallabhbhai Patel
(D) Jawaharlal Nehru

Q.11
Akbar's first expedition towards Deccan was sent to Ahmednagar under the leadership of Prince Murad and _____.

(A) Adham Khan
(B) Man Singh
(C) Birbal
(D) Abdul Rahim Khan-i-Khanan

Q.12
Which Rigvedic God is considered an intermediary between God and men?

(A) Varuna
(B) Indra
(C) Agni
(D) Yama

Q.13
How many central banks can a country have?

(A) One
(B) Two
(C) Three
(D) Four

Q.14
Which of the following is written by Mulk Raj Anand?

(A) Untouchable
(B) The Guide
(C) The Blue Umbrella
(D) The White Tiger

Q.15
The Uttarakhand Public Services (Horizontal Reservation for Women) Bill, 2022 provides how much reservation to women?

(A) 20%
(B) 30%
(C) 25%
(D) 45%

Q.16
A regal ceremony in the form of Beating Retreat ceremony on the evening of _____ marks the end of Republic Day celebrations.

(A) 26th January
(B) 28th January
(C) 27th January
(D) 29th January

Q.17
What is the rate of movement of tectonic plate boundaries?

(A) 1-2 m per year
(B) 1-2 inch per year
(C) 2-5 km per year
(D) 1-2 mm per year

Q.18
Which of the following is NOT mentioned in 'Liberty' as per the Preamble of the Indian Constitution?

(A) Faith and worship
(B) Expression
(C) Thought
(D) Trust and hope

Q.19
Based on the number of chambers in the heart, select the odd one from the following.

(A) Frog
(B) Chameleon
(C) Turtle
(D) Crocodile

Q.21
Padma Subramanyam, Alarmel Valli, Yamini Krishnamurthy and Anita Ratnam are distinguished practitioners of _____.

(A) Bharatanatyam
(B) Odissi
(C) Kuchipudi
(D) Kathak

Q.22
Which of the following Biosphere reserves is located in Uttarakhand?

(A) Panchmarhi
(B) Nilgiri
(C) Nanda Devi
(D) Dihang-Dibang

Q.23
In February 2023, who was appointed as the CEO of NITI Aayog?

(A) BVR Subrahmanyam
(B) Iyer Parmeshwaran
(C) Geetika Srivastava
(D) Monali Jaiswal

Q.24
In which state is the Kaiga Nuclear Power Project located?

(A) Assam
(B) Bihar
(C) Gujarat
(D) Karnataka

Q.25
Theravada is known as the official religion of which of the following neighbouring countries of India?

(A) Bangladesh
(B) Afghanistan
(C) Sri Lanka
(D) Pakistan

English Language and Comprehension 25 questions

Q.1
Your sibling accidentally broke your favourite vase. Which of the following idioms might you use to tell them not to worry about it?

(A) "Shallow brooks are noisy"
(B) "You're walking on air!"
(C) "Bite the bullet!"
(D) "Don't cry over spilt milk!"

Q.2
Select the most appropriate option to fill in the blank. The humble surroundings of our company _____ creativity amongst the newly recruited employees.

(A) banished
(B) inspired
(C) connected
(D) filed

Q.3

Select the option that can be used as a one-word substitute for the given group of words. The act of talking to oneself, often audibly

(A) Dialogue
(B) Soliloquy
(C) Solitude
(D) Monologue

Q.4

Select the INCORRECTLY spelt word.

(A) Overview
(B) Oblegation
(C) Ointment
(D) Opponent

Q.5

Select the most appropriate ANTONYM of the given word. Fetid

(A) Fragrant
(B) Fatal
(C) False
(D) Fatigued

Q.6

Select the option that will improve the underlined part of the given sentence. We agree that Professor Revanth is very knowledgeable, but we do not like attending his classes because he always keeps “blowing the trumpet” .

(A) blowing his own trumpet
(B) the blow of a trumpet
(C) sounding the trumpet
(D) the sound of the trumpet

Q.7

Select the option that will correct the underlined part of the given sentence. In case no correction is required, select No correction needed. Naveen is “as tall as” Rocky.

(A) No correction needed
(B) tall as
(C) the tallest
(D) taller than

Q.8

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He hardness ever / complains about / his workload because / he enjoys his job.

(A) his workload because
(B) complains about
(C) He hardness ever
(D) he enjoys his job

Q.9

Select the most appropriate synonym of the underlined word in the given sentence. I have a “vague” memory of meeting her many years ago.

(A) evident
(B) pellucid
(C) Symbolic
(D) Hazy

Q.10

Select the most appropriate word opposite in meaning (ANTONYM) to the underlined word in the given sentence. Aunty Hema gave us her “candid” opinion on the skit we performed on Children’s Day.

(A) frank
(B) prejudiced
(C) strong
(D) intelligent

Q.11

Select the most appropriate option to fill in the blank. I dream of a country where even the least talented person has the _____ to earn a living.

(A) priority
(B) admit
(C) landmark
(D) opportunity

Q.12
Select the most appropriate ANTONYM of the underlined word in the given sentence. The late secretary of our society was known for his “ruthless” treatment of all the security personnel of the society.

(A) Fragile
(B) Compassionate
(C) Brutal
(D) Startled

Q.13
Select the INCORRECTLY spelt word. A) Exaggerate B) Accommodate C) Exuberant D) Perservere

(A) B
(B) C
(C) A
(D) D

Q.14
Parts of the following sentence have been given as options. Select the option that contains an error. An eucalyptus tree in my backyard acts as a landmark for many to reach my house.

(A) An eucalyptus tree
(B) acts as a landmark for many
(C) to reach my house
(D) in my backyard

Q.15
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. How can someone / walk this farthest / just for / peace of mind?

(A) How can someone
(B) peace of mind
(C) walk this farthest
(D) just for

Q.16
Select the option that corrects the error in the given sentence. She performed bad in the exam.

(A) goodly
(B) badly
(C) bad
(D) good

Q.17
Select the most appropriate ANTONYM of the given word. Show

(A) Exhibit
(B) Register
(C) Display
(D) Cover

Q.18
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The fresh air / in the garden relaxes / the minds of children as / the environment is cleaner.

(A) the environment is cleaner
(B) the minds of children as
(C) in the garden relaxes
(D) The fresh air

Q.19
Select the INCORRECTLY spelt word in the given sentence. Applicants who are younger and prettier than you will continue appliying.

(A) younger
(B) applicants
(C) prettier
(D) appliying

Q.20
Select the most appropriate option to fill in the blank. The teacher asked us to _____ our essays for spelling and grammar errors.

(A) advise
(B) recognise
(C) revise
(D) revolve

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Students want (1)_____ as the primary outcome of higher education. Hence, practical experience becomes (2)_____ valuable to a student's degree. Indian students are highly (3)_____ at theoretical learning but lack practical training. This (4)_____ an issue for decades in the Indian job market. In order to address this (5)_____ issue, state governments have to take a lead. The AP government in the past three years have made efforts to bridge this long-lasting gap between theoretical and practical learning of the students.

Select the most appropriate option to fill in blank 1.

(A) employability
(B) appointed
(C) employable
(D) appoint

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Students want (1)_____ as the primary outcome of higher education. Hence, practical experience becomes (2)_____ valuable to a student's degree. Indian students are highly (3)_____ at theoretical learning but lack practical training. This (4)_____ an issue for decades in the Indian job market. In order to address this (5)_____ issue, state governments have to take a lead. The AP government in the past three years have made efforts to bridge this long-lasting gap between theoretical and practical learning of the students.

Select the most appropriate option to fill in blank 2.

(A) abruptly
(B) equally
(C) voicelessly
(D) hardly

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Students want (1)_____ as the primary outcome of higher education. Hence, practical experience becomes (2)_____ valuable to a student's degree. Indian students are highly (3)_____ at theoretical learning but lack practical training. This (4)_____ an issue for decades in the Indian job market. In order to address this (5)_____ issue, state governments have to take a lead. The AP government in the past three years have made efforts to bridge this long-lasting gap between theoretical and practical learning of the students.

Select the most appropriate option to fill in blank 3.

(A) spiritual
(B) spirit
(C) relaxed
(D) competitive

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Students want (1)_____ as the primary outcome of higher education. Hence, practical experience becomes (2)_____ valuable to a student's degree. Indian students are highly (3)_____ at theoretical learning but lack practical training. This (4)_____ an issue for decades in the Indian job market. In order to address this (5)_____ issue, state governments have to take a lead. The AP government in the past three years have made efforts to bridge this long-lasting gap between theoretical and practical learning of the students.

Select the most appropriate option to fill in blank 4.

(A) is
(B) have been
(C) will be
(D) has been

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Students want (1)_____ as the primary outcome of higher education. Hence, practical experience becomes (2)_____ valuable to a student's degree. Indian students are highly (3)_____ at theoretical learning but lack practical training. This (4)_____ an issue for decades in the Indian job market. In order to address this (5)_____ issue, state governments have to take a lead. The AP government in the past three years have made efforts to bridge this long-lasting gap between theoretical and practical learning of the students.

Select the most appropriate option to fill in blank 5.

(A) insignificant

(B) momentary

(C) perennial

(D) appetite

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(B)**	Q.8 **(D)**	Q.9 **(B)**
Q.10 **(C)**	Q.11 **(A)**	Q.12 **(C)**	Q.14 **(C)**	Q.15 **(D)**	Q.16 **(C)**
Q.19 **(A)**	Q.20 **(B)**				

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.2 **(A)**	Q.4 **(C)**	Q.6 **(B)**	Q.7 **(C)**	Q.8 **(C)**
Q.9 **(A)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(C)**	Q.14 **(C)**	Q.16 **(D)**
Q.17 **(C)**	Q.18 **(C)**	Q.19 **(A)**			

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(C)**	Q.9 **(D)**	Q.10 **(B)**	Q.11 **(D)**	Q.12 **(C)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(B)**	Q.16 **(D)**	Q.17 **(B)**	Q.18 **(D)**
Q.19 **(D)**	Q.21 **(A)**	Q.22 **(C)**	Q.23 **(A)**	Q.24 **(D)**	Q.25 **(C)**

English Language and Comprehension

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(A)**
Q.7 **(A)**	Q.8 **(C)**	Q.9 **(D)**	Q.10 **(B)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(D)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(B)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(A)**	Q.22 **(B)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(C)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 36,000

P × 1.05 × 0.94 = 0.987P = 35,532, so P = 36,000.

Q.3 · Answer (B) 66

The LCM of 20, 15 and 30 is 60, so the smallest such number is 60 + 6 = 66.

Q.4 · Answer (B) 2.23

The 8 litres hold 3.2 litres of antifreeze. Adding x litres of 95%: 3.2 + 0.95x = 0.52(8 + x), so 0.43x = 0.96 and x ≈ 2.23 litres.

Q.5 · Answer (B) 450

SI = 3,000 × 3 × 5/100 = ₹450.

Q.8 · Answer (D) 20 minutes

4 minutes 10 seconds is 250 seconds for 2.5 km. 12 km is 4.8 times as far: 4.8 × 250 = 1,200 seconds = 20 minutes.

Q.9 · Answer (B) 49.8

(40 + 49 + 56 + 69 + 35) ÷ 5 = 249 ÷ 5 = 49.8 kg.

Q.10 · Answer (C) 5

Outer radius R = 565.2 ÷ (2 × 3.14) = 90 cm. TSA = outer $2\pi R^2$ + inner $2\pi r^2$ + rim $\pi(R^2 - r^2)$ = $\pi(3R^2 + r^2)$, so $3.14(24{,}300 + r^2)$ = 98,988.5, r^2 = 7,225 and r = 85 cm. y = 90 − 85 = 5 cm.

Q.11 · Answer (A) 30 years

Doubling means the interest equals the sum in 10 years, so the rate is 10%. To become 4 times, the interest must be 3 times the sum: 30 years.

Q.12 · Answer (C) 450

SI = 3,000 × 5 × 3/100 = ₹450.

Q.14 · Answer (C) 60 km/h

Time = 120/48 + 120/80 = 2.5 + 1.5 = 4 hours for 240 km: 240 ÷ 4 = 60 km/h.

Q.15 · Answer (D) 25 : 48

SP = 0.75 × printed price = 1.44 × CP, so CP : printed price = 0.75 : 1.44 = 25 : 48.

Q.16 · Answer (C) 80.75

Total marks = 30 × 90 + 36 × 85 + 40 × 70 = 2,700 + 3,060 + 2,800 = 8,560 for 106 students: 8,560 ÷ 106 = 80.75.

Q.19 · Answer (A) 192

Equal resistance means L/r^2 is the same for both wires: L = 147 × $(8/7)^2$ = 147 × 64/49 = 192 cm.

Q.20 · Answer (B) ₹247

(3 × 240 + 1 × 258 + 2 × 252) ÷ 6 = (720 + 258 + 504) ÷ 6 = 1,482 ÷ 6 = ₹247.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) L

Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise from F the seats are F, K, E, H, L, G, M (the other place for E and H puts H next to F). L sits to G's immediate right.

Q.2 · Answer (A) 22

Each code is twice the number of letters plus 4: STATION has 7 letters (2 × 7 + 4 = 18) and JOURNALS 8 (2 × 8 + 4 = 20). GOLDMEDAL has 9: 2 × 9 + 4 = 22.

Q.4 · Answer (C) system
'is' and 'backbone' are in both sentences, as are 'wer' and 'tce' in both codes. The word left in the first sentence is 'system', and the code left is 'sdf'.

Q.6 · Answer (B) 39
After interchanging the signs: 35 – 4 × 6 ÷ 3 + 12 = 35 – 8 + 12 = 39.

Q.7 · Answer (C) Neither conclusion I nor II follows
All roses being lilies does not make all lilies roses (I). The lilies that are not roses may be sunflowers, so II does not follow either.

Q.8 · Answer (C) Both conclusions (I) and (II) follow.
Every duffle is a container, so some containers are duffles (I); bags are duffles and duffles are containers, so all bags are containers (II). Both follow.

Q.9 · Answer (A) 13
After interchanging the signs: 12 ÷ 3 + 5 × 4 – 11 = 4 + 20 – 11 = 13.

Q.10 · Answer (D) 3
W is the only letter in both words and 3 the only digit in both codes, so W = 3.

Q.11 · Answer (D) 0
Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise from C the seats are C, D, F, E, A, G, B, the only order that meets every condition. B sits immediately to G's left, so no one is between them.

Q.12 · Answer (C) 76
The numbers rise by 7 each time: 69 + 7 = 76.

Q.14 · Answer (C) ALRN
The first two letters move 5 places forward and the last two 3 back, wrapping round the alphabet: BMGC → GRDZ and LWAW → QBXT. So VGUQ → ALRN.

Q.16 · Answer (D) 186
The second number is ten times the first minus 4: 11 × 10 – 4 = 106 and 15 × 10 – 4 = 146. So 19 × 10 – 4 = 186.

Q.17 · Answer (C) VT57
Both letters move 4 places forward and the number rises by 5: MK14 → QO19 and XV31 → BZ36 (X wraps round to B). So RP52 → VT57.

Q.18 · Answer (C) TRV
Each letter moves 4 places forward from one group to the next: D, H, L, P, T; B, F, J, N, R; F, J, N, R, V. The next group is TRV.

Q.19 · Answer (A) (212, 166)
The second number is 46 less than the first: 284 – 46 = 238 and 271 – 46 = 225. Likewise 212 – 46 = 166.

General Awareness

Q.1 · Answer (C) Sasaram
Sher Shah Suri's tomb stands in an artificial lake at Sasaram, in Bihar's Rohtas district.

Q.2 · Answer (D) Mir Jafar Ali Khan
After Plassey the British made Mir Jafar, Siraj-ud-Daulah's commander who had gone over to them, the Nawab of Bengal.

Q.3 · Answer (D) India
India beat Sri Lanka by ten wickets in the final at Colombo in September 2023 for its eighth Asia Cup title.

Q.4 · Answer (A) Abhinandan Saroja
SSC's key gives the Abhinandan Saroja National Award, which the National Institute of Indian Classical Dance presents to classical dancers.

Q.5 · Answer (B) Carnatic music
Ranjani Hebbar (died 2013) was a Carnatic vocalist from Udupi, Karnataka, and an A-grade artist of All India Radio.

Q.6 · Answer (C) Fluorine
Fluorine is the most electronegative element, at 3.98 on the Pauling scale.

Q.7 · Answer (C) A tied score at 40-40
Deuce is a score of 40-40 in a game of tennis; a player must then win two points in a row to take the game.

Q.8 · Answer (C) Rhinovirus
SSC's key gives rhinovirus, the cause of most common colds. Some coronaviruses also cause colds, though far fewer; influenza viruses cause flu, and hepatitis A a liver infection.

Q.9 · Answer (D) 2002
The 86th Amendment Act, 2002 added Article 51A(k), the duty of parents or guardians to provide opportunities for education to their children aged 6 to 14.

Q.10 · Answer (B) Mahatma Gandhi
Mahatma Gandhi led the Salt Satyagraha, marching from Sabarmati to Dandi in March-April 1930 to break the salt law.

Q.11 · Answer (D) Abdul Rahim Khan-i-Khanan
Akbar's first Deccan campaign, against Ahmednagar, was led by Prince Murad and Abdul Rahim Khan-i-Khanan; Chand Bibi defended the city.

Q.12 · Answer (C) Agni
In the Rig Veda, Agni, the god of fire, carries the sacrificial offerings to the gods, and so acts as the messenger between gods and men.

Q.13 · Answer (A) One
A country has a single central bank, which issues its currency and regulates its banks; India's is the Reserve Bank of India.

Q.14 · Answer (A) Untouchable
'Untouchable' (1935) is Mulk Raj Anand's first novel. 'The Guide' is by R. K. Narayan, 'The Blue Umbrella' by Ruskin Bond and 'The White Tiger' by Aravind Adiga.

Q.15 · Answer (B) 30%
The Uttarakhand bill of 2022 gives women of Uttarakhand domicile 30% horizontal reservation in the state's public services.

Q.16 · Answer (D) 29th January
The Beating Retreat ceremony at Vijay Chowk, New Delhi, on the evening of 29 January closes the Republic Day celebrations.

Q.17 · Answer (B) 1-2 inch per year
Tectonic plates move only a few centimetres a year; of these options, 1-2 inches (about 2.5-5 cm) a year.

Q.18 · Answer (D) Trust and hope
The Preamble promises liberty of thought, expression, belief, faith and worship; 'trust and hope' are not among them.

Q.19 · Answer (D) Crocodile
Frogs, chameleons and turtles have three-chambered hearts; the crocodile, like the other crocodilians (alone among reptiles), has four.

Q.21 · Answer (A) Bharatanatyam
Padma Subrahmanyam, Alarmel Valli, Yamini Krishnamurthy and Anita Ratnam are all noted Bharatanatyam dancers (Yamini Krishnamurthy was also a leading Kuchipudi dancer).

Q.22 · Answer (C) Nanda Devi
The Nanda Devi Biosphere Reserve is in Uttarakhand. Pachmarhi is in Madhya Pradesh, the Nilgiri in Tamil Nadu, Kerala and Karnataka, and Dihang-Dibang in Arunachal Pradesh.

Q.23 · Answer (A) BVR Subrahmanyam
BVR Subrahmanyam was appointed CEO of NITI Aayog in February 2023, succeeding Parameswaran Iyer.

Q.24 · Answer (D) Karnataka
The Kaiga nuclear power station is in Uttara Kannada district, Karnataka.

Q.25 · Answer (C) Sri Lanka
Theravada Buddhism is the main religion of Sri Lanka, whose constitution gives Buddhism 'the foremost place'. Bangladesh, Afghanistan and Pakistan are Muslim-majority countries.

English Language and Comprehension

Q.1 · Answer (D) "Don’t cry over spilt milk!"
'Don't cry over spilt milk' means there is no point in worrying about something that has already happened and cannot be undone.

Q.2 · Answer (B) inspired
The surroundings 'inspired', that is encouraged, creativity; 'banished' says the opposite, and 'connected' and 'filed' do not fit.

Q.3 · Answer (B) Soliloquy
A soliloquy is the act of speaking one's thoughts aloud to oneself; a monologue is a long speech by one person to others.

Q.4 · Answer (B) Oblegation
The correct spelling is 'obligation'.

Q.5 · Answer (A) Fragrant
'Fetid' means foul-smelling; its opposite is 'fragrant'.

Q.6 · Answer (A) blowing his own trumpet
To 'blow one's own trumpet' is to boast about oneself.

Q.7 · Answer (A) No correction needed
Equality is shown with 'as … as' and the positive degree, so the sentence is correct as it stands.

Q.8 · Answer (C) He hardness ever
'Hardly ever' means almost never: 'He hardly ever complains about his workload'. 'Hardness' is a noun.

Q.9 · Answer (D) Hazy
'Vague' means unclear or indistinct: 'hazy'.

Q.10 · Answer (B) prejudiced
'Candid' means frank and, in an older sense, free from bias; its opposite here is 'prejudiced'. 'Frank' is a synonym.

Q.11 · Answer (D) opportunity
Even the least talented person should have the 'opportunity', the chance, to earn a living.

Q.12 · Answer (B) Compassionate
'Ruthless' means without pity; its opposite is 'compassionate'.

Q.13 · Answer (D) D
The correct spelling is 'persevere'.

Q.14 · Answer (A) An eucalyptus tree
'Eucalyptus' begins with a 'yoo' sound, a consonant sound, so it takes 'a': 'A eucalyptus tree'.

Q.15 · Answer (C) walk this farthest
'This far' means to such a distance: 'walk this far'. The superlative 'farthest' does not go with 'this'.

Q.16 · Answer (B) badly
A verb is modified by an adverb: 'She performed badly in the exam.'

Q.17 · Answer (D) Cover
The opposite of 'show' is 'cover', to hide from view; 'exhibit' and 'display' are synonyms.

Q.18 · Answer (A) the environment is cleaner
Nothing is being compared, so the positive degree: 'as the environment is clean'.

Q.19 · Answer (D) appliying
The correct spelling is 'applying'.

Q.20 · Answer (C) revise
To 'revise' an essay is to read it again and correct it.

Q.21 · Answer (A) employability
Students want 'employability', a noun, as the outcome of higher education; 'employable' is an adjective.

Q.22 · Answer (B) equally
Practical experience becomes 'equally' valuable, as valuable as the degree itself.

Q.23 · Answer (D) competitive
Indian students are 'highly competitive' at theoretical learning, in contrast with their lack of practical training.

Q.24 · Answer (D) has been
'For decades' calls for the present perfect: 'This has been an issue'.

Q.25 · Answer (C) perennial
A 'perennial' issue is one that lasts or keeps returning: the 'long-lasting gap' the passage goes on to describe.

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